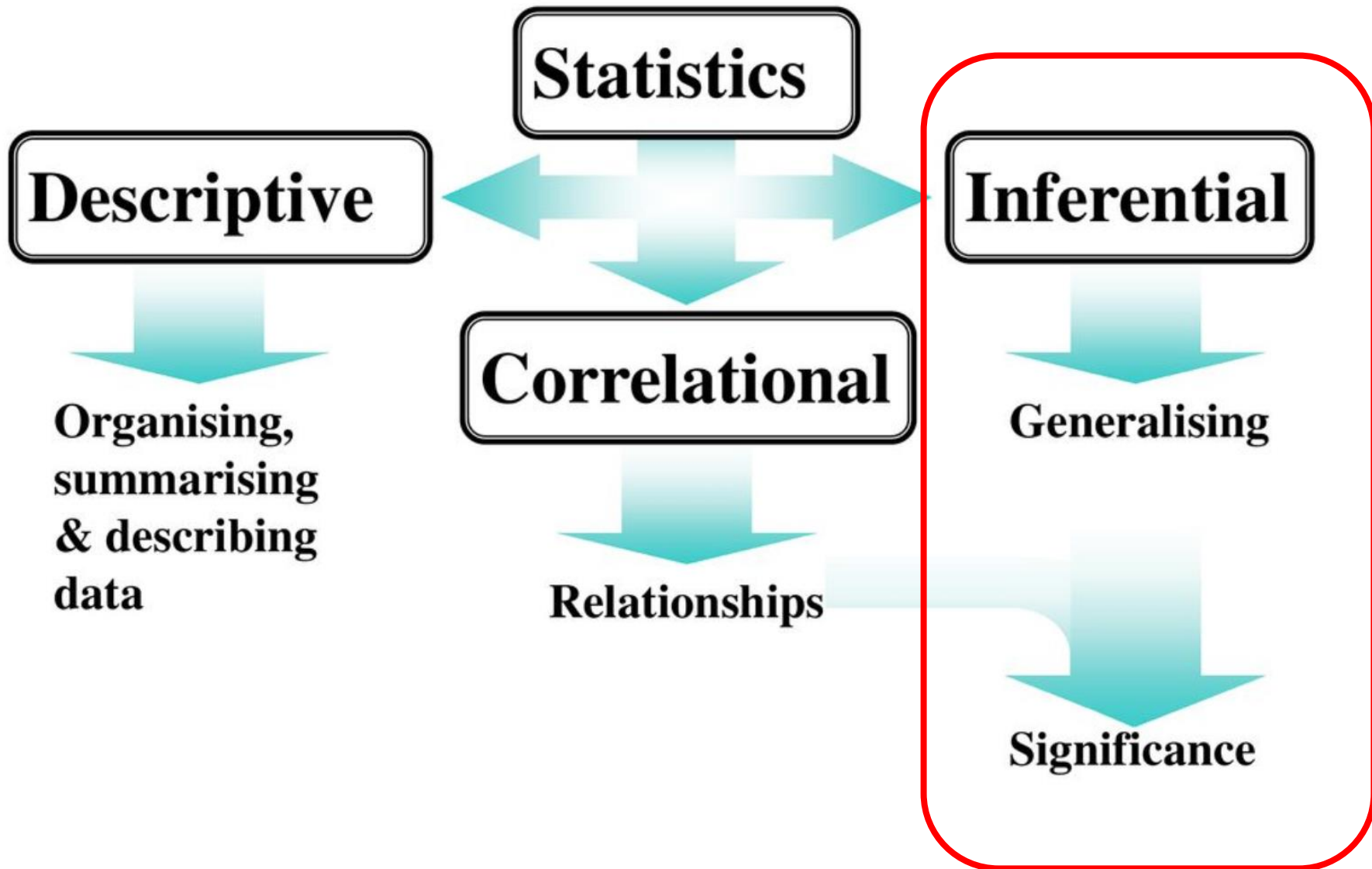


Non-parametric tests

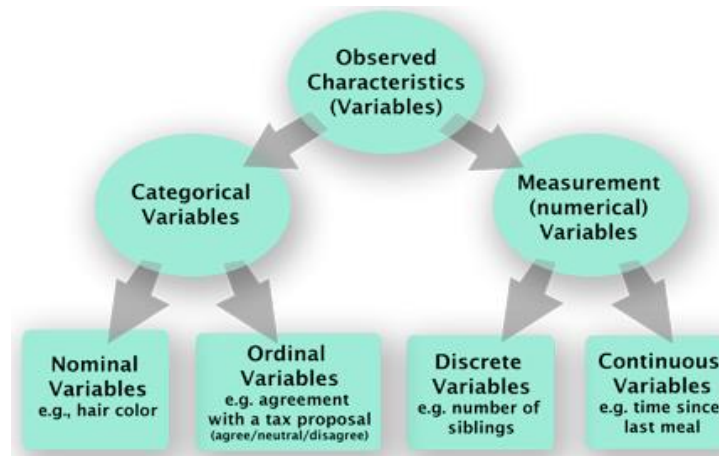
How to deal with Categorical data?

How to deal with cases where parametric assumptions are violated?

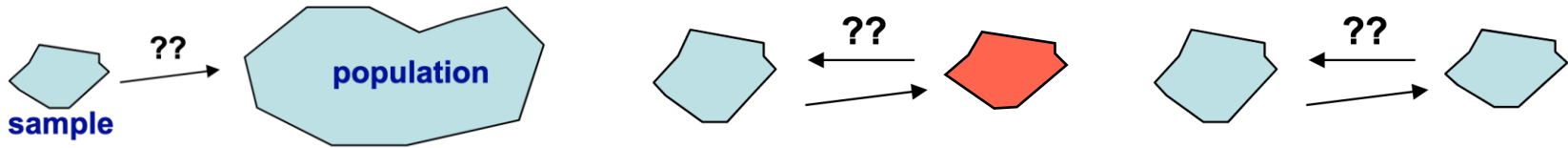


Selecting a statistical test

- Different tests are used according to the level of measurement:
 - Interval
 - Ordinal
 - Categorical
- Parametric vs non-parametric (makes no assumption on the population distribution or sample size) assumptions



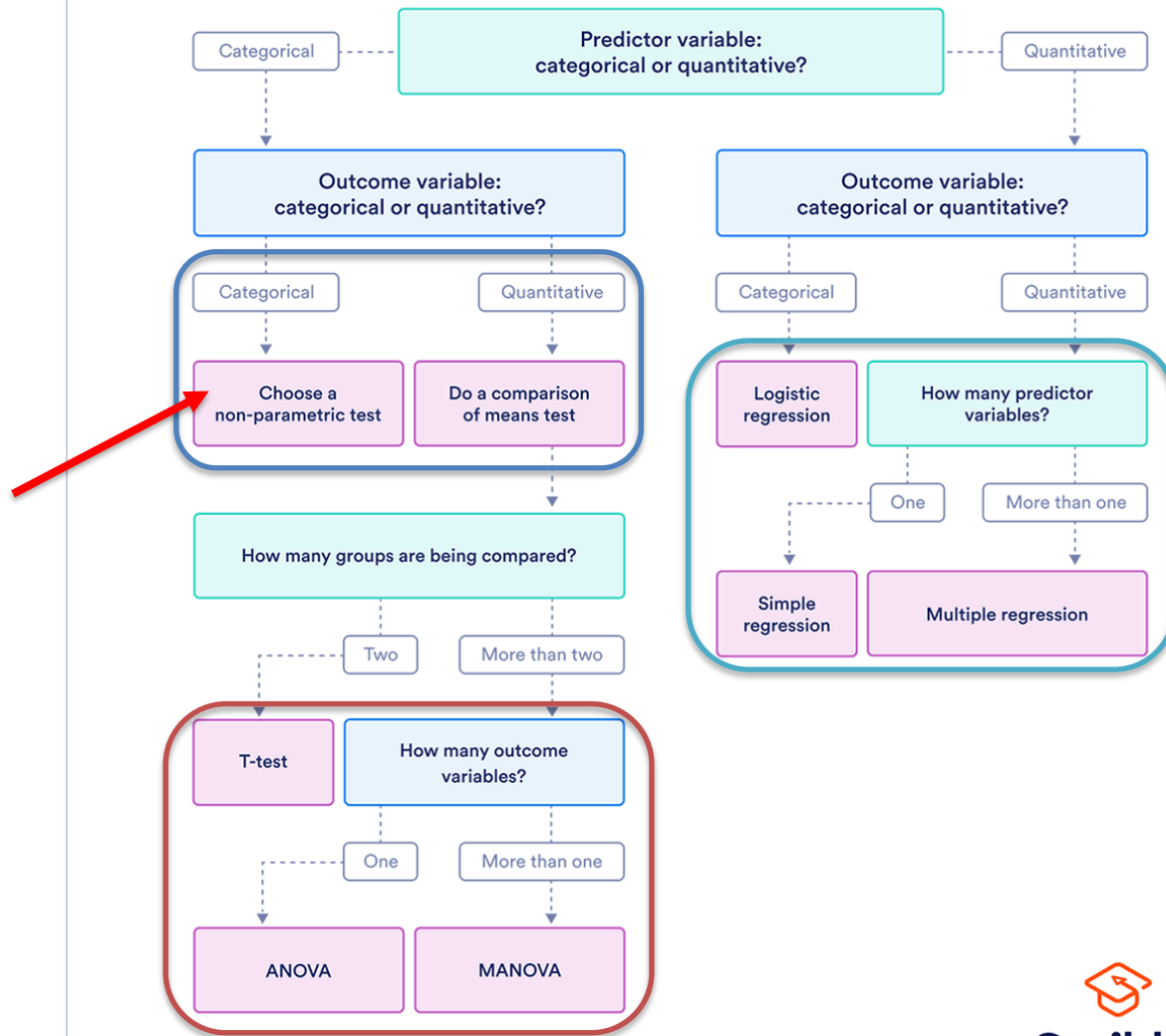
Selecting a statistical test



- Different tests are used for varying amount of groups/conditions:
 - two samples
 - > two samples
- Different tests are used for related versus unrelated designs:
 - unrelated samples = between subjects designs
 - related samples = within subjects designs & matched pairs

Choosing a statistical test

This flowchart helps you choose among parametric tests



	Predictor variable	Outcome variable	Research question example
Simple linear regression	- Continuous 1 predictor	- Continuous 1 outcome	What is the effect of income on longevity?
Multiple linear regression	- Continuous 2 or more predictors	- Continuous 1 outcome	What is the effect of income and minutes of exercise per day on longevity?
Logistic regression	Continuous	Binary	What is the effect of drug dosage on the survival of a test subject?

Regression tests

Regression tests look for **cause-and-effect** relationships. They can be used to estimate the effect of one or more continuous variables on another variable.

	Predictor variable	Outcome variable	Research question example
Paired t-test	• Categorical • 1 predictor	• Quantitative • groups come from the same population	What is the effect of two different test prep programs on the average exam scores for students from the same class?
Independent t-test	• Categorical • 1 predictor	• Quantitative • groups come from different populations	What is the difference in average exam scores for students from two different schools?
ANOVA	• Categorical • 1 or more predictor	• Quantitative • 1 outcome	What is the difference in average pain levels among post-surgical patients given three different painkillers?
MANOVA	• Categorical • 1 or more predictor	• Quantitative • 2 or more outcome	What is the effect of flower species on petal length, petal width, and stem length?

Comparison tests

Comparison tests look for **differences among group means**. They can be used to test the effect of a categorical variable on the mean value of some other characteristic.

T-tests are used when comparing the means of precisely two groups (e.g., the average heights of men and women). ANOVA and MANOVA tests are used when comparing the means of more than two groups (e.g., the average heights of children, teenagers, and adults).

Correlation tests

Correlation tests **check whether variables are related** without hypothesizing a cause-and-effect relationship.

These can be used to test whether two variables you want to use in (for example) a multiple regression test are autocorrelated.

	Variables	Research question example
Pearson's r	• 2 continuous variables	How are latitude and temperature related?

Selecting a statistical test

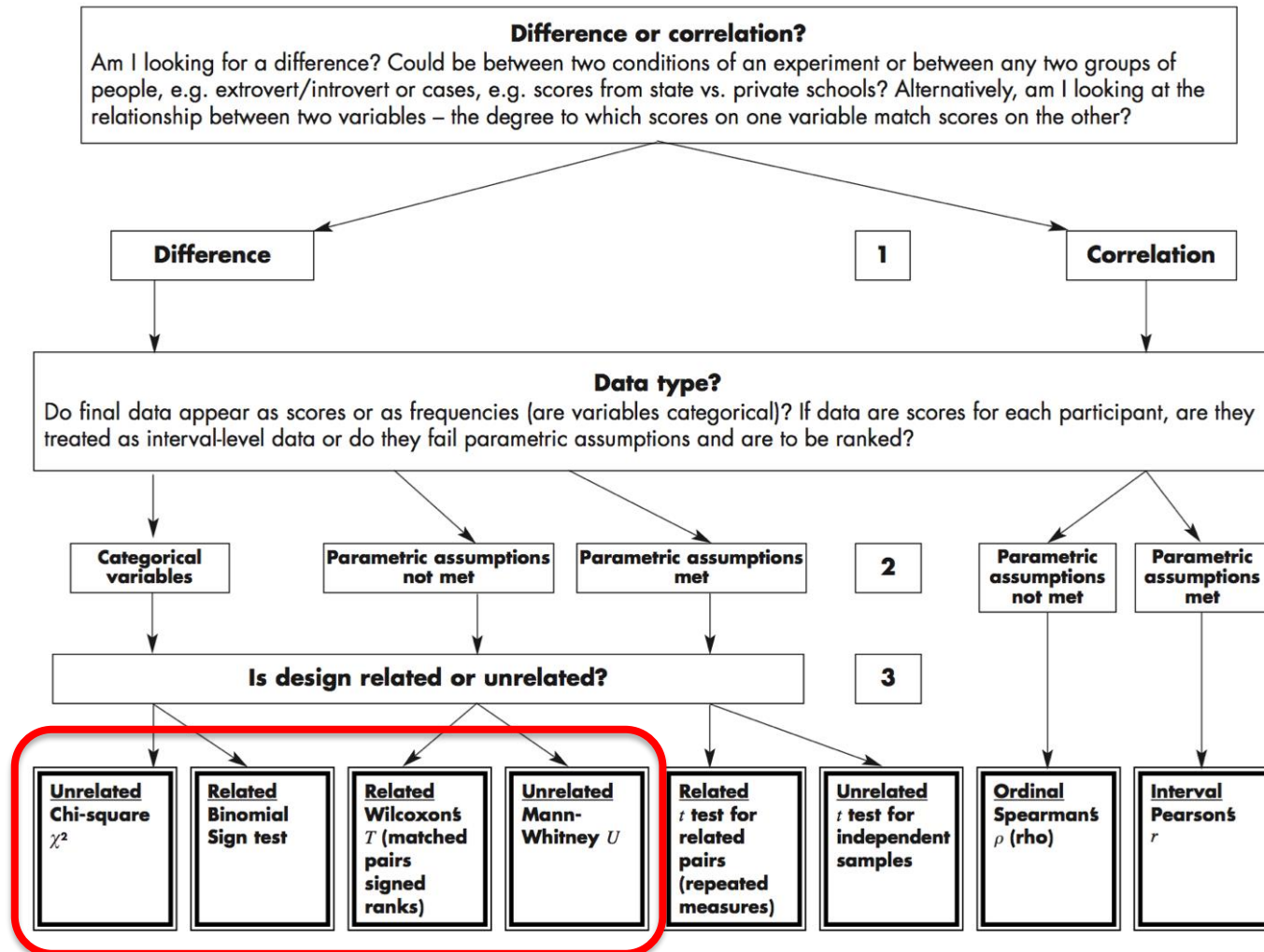


Figure 23.1 Choosing an appropriate two-sample test.

	Predictor variable	Outcome variable	Use in place of...
Spearman's r	Quantitative	Quantitative	Pearson's r
Chi square test of independence	Categorical	Categorical	Pearson's r
Sign test	Categorical	Quantitative	One-sample t -test
Kruskal-Wallis H	- Categorical 3 or more groups	Quantitative	ANOVA
ANOSIM	- Categorical 3 or more groups	- Quantitative 2 or more outcome variables	MANOVA
Wilcoxon Rank-Sum test	- Categorical 2 groups	- Quantitative - groups come from	Independent t -test
Wilcoxon Signed-rank test	- Categorical 2 groups	- Quantitative - groups come from the same population	Paired t -test

Choosing a nonparametric test

Non-parametric tests don't make as many assumptions about the data, and are useful when one or more of the common statistical assumptions are violated. However, the inferences they make aren't as strong as with parametric tests.

Different types of tests

Test type	Between subjects designs (Independent samples)	Within subject designs (repeated measures/matched pairs)
Non-parametric (for categorical data)	<i>Chi-square test</i>	<i>The binomial sign test</i>
Non-parametric (for ordinal data)	<i>Mann-Whitney U</i>	<i>Wilcoxon Signed-Rank Test</i> <i>The binomial sign test</i>
Parametric	<i>Unrelated t-test (level of data: interval)</i>	<i>Related t-test (level of data: interval)</i>

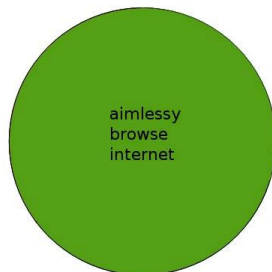
Chi-Square Test

Theoretical
categorical distribution
vs
Observed
categorical distribution

Weekend in college



Expectation



Reality

preference for one brand

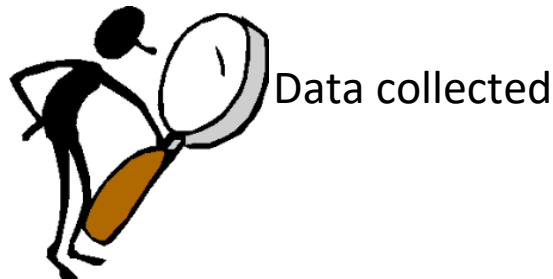
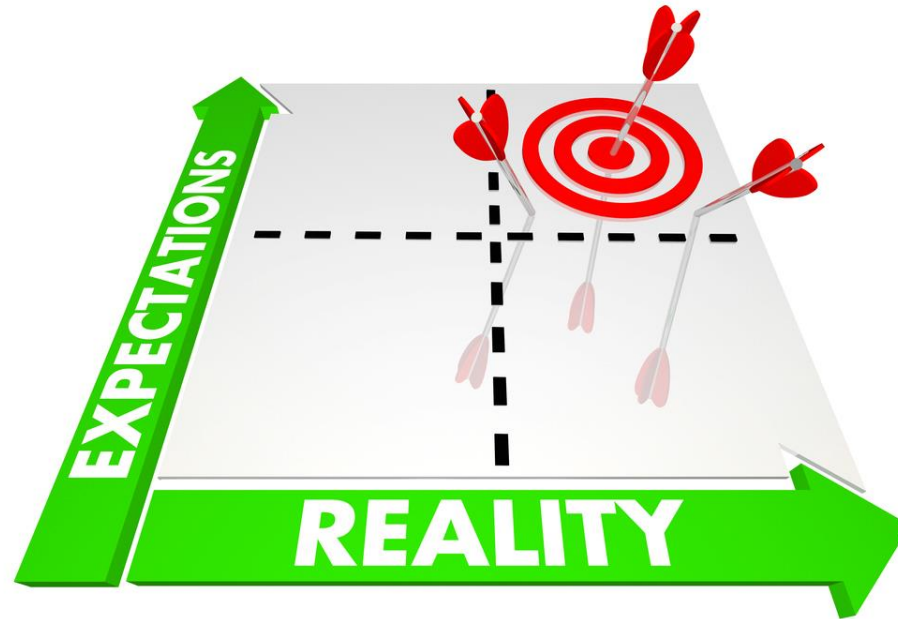


Chi-Square test

$$\chi^2_c = \sum \frac{(O_i - E_i)^2}{E_i}$$

H₀

H_A



Chi-Square Test - Applications

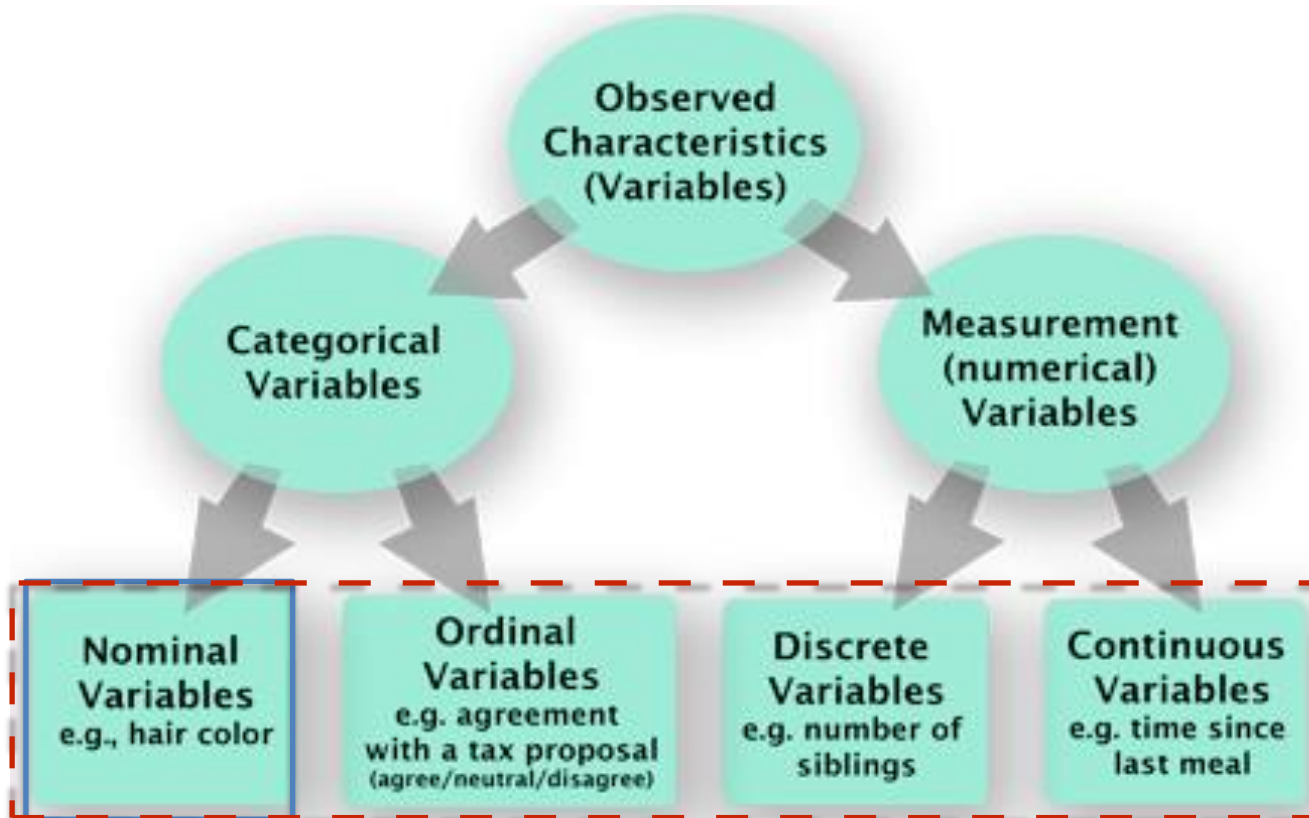
- Goodness-of-fit:
 - compare the observed sample distribution with the expected probability distribution
 - H_0 = no difference from a known population
- Chi-Square fit test:
 - determines how well theoretical distribution fits the empirical distribution
 - H_0 = no difference, equal proportions

Chi-Square Test - Applications

- Test for Independence (for two variables):
 - test *relationship* between two separate variables
 - H_0 = there is no relationship between the variables
 - eg: females prefer pepsi more than males



Chi-Square Test



Applications of Chi-square test

- Chi-square test is used to compare categorical variables. There are two type of chi-square test
 - 1. **Goodness of fit test** which determines if a sample matches the population.
 - 2. A chi-square fit **test for two independent variables** is used to compare two variables in a contingency table to check if the data fits.
 - The hypothesis being tested for chi-square is
 - Null: Variable A and Variable B are independent
 - Alternate: Variable A and Variable B are not independent.
- What to expect?
 - A small chi-square value means that data fits / variables independent
 - A high chi-square value means that data doesn't fit / variables not independent

$$\chi^2_c = \sum \frac{(O_i - E_i)^2}{E_i}$$



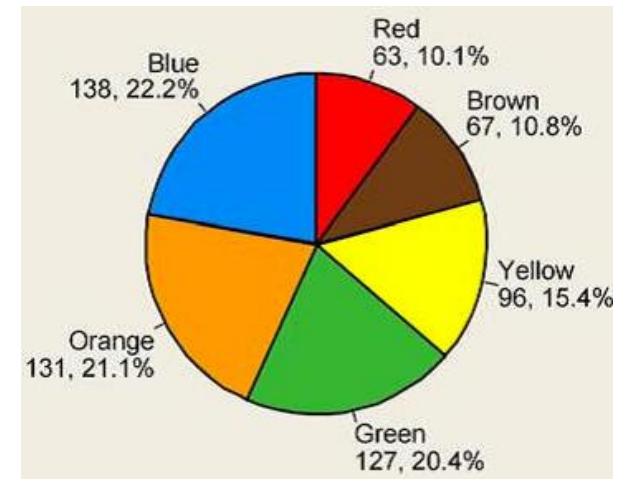
Chi-Square test/goodness-of-fit

EXAMPLE



H_0

M&Ms Color Distribution %
according to their website



H_0 : The color distribution is equal

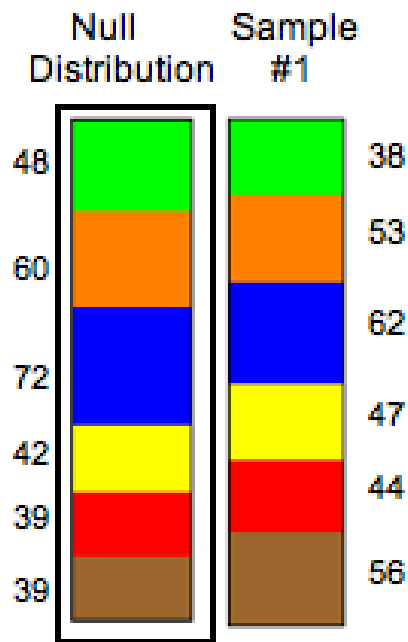
revised H_0 :

The color distribution is 13% brown, 13% red, 14% yellow, 24% blue, 20% orange, 16% green

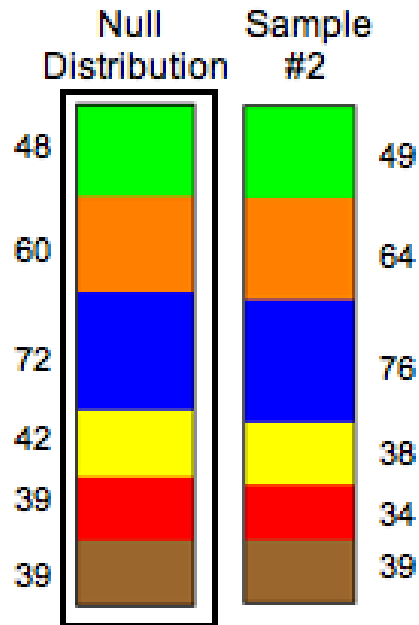
H_A : The color distribution is different from 13% brown, 13% red, 14% yellow, 24% blue, 20% orange, 16% green



$$\chi^2_c = \sum \frac{(O_i - E_i)^2}{E_i}$$



$$\chi^2 = 12.94$$



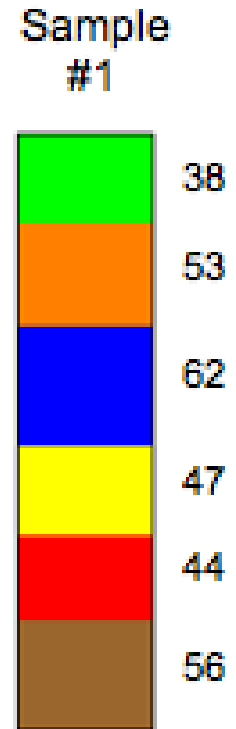
$$\chi^2 = 1.53$$

df = ?

H₀ ?

Chi-Square distribution & df

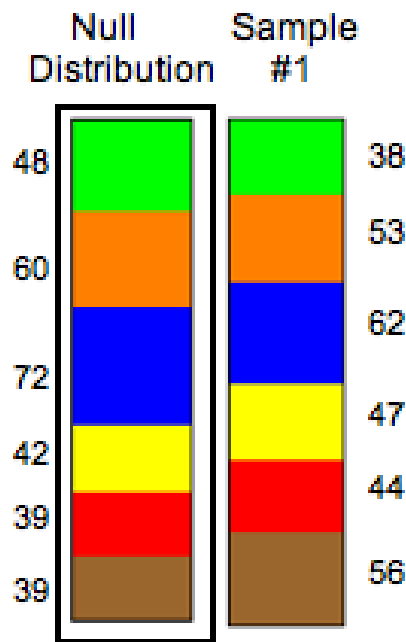
- Degrees of freedom for goodness-of-fit
 - number of cells you would need to calculate all other cell values, assuming we know marginal values
- $df = C - 1$, C = no. of categories



df = ?

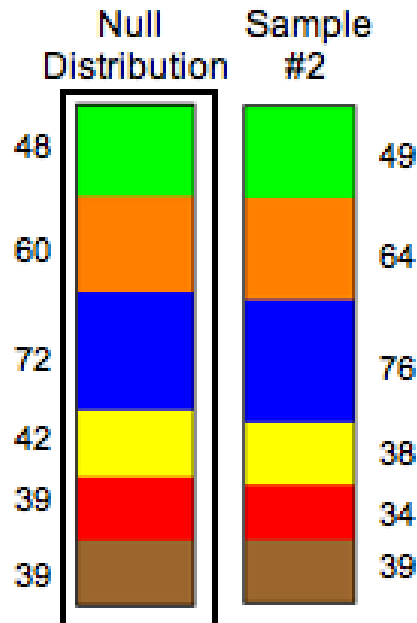


$$\chi^2_c = \sum \frac{(O_i - E_i)^2}{E_i}$$



$$\chi^2 = 12.94$$

REJECTED H_0



$$\chi^2 = 1.53$$

? **ACCEPTED**

df = 5

Critical values of the Chi-square distribution with d degrees of freedom

Probability of exceeding the critical value						
d	0.05	0.01	0.001	d	0.05	0.01
1	3.841	6.635	10.828	11	19.675	24.725
2	5.991	9.210	13.816	12	21.026	26.217
3	7.815	11.345	16.266	13	22.362	27.688
4	9.488	13.277	18.467	14	23.685	29.141
5	11.070	15.086	20.515	15	24.996	30.578
6	12.592	16.812	22.458	16	26.296	32.000
7	14.067	18.475	24.322	17	27.587	33.409
8	15.507	20.090	26.125	18	28.869	34.805
9	16.919	21.666	27.877	19	30.144	36.191
10	18.307	23.209	29.588	20	31.410	37.566

Degrees of Freedom (*df*)

- number of independent pieces of information that go into the estimate of a parameter
- *df* depends on
 - particular calculation you will be performing
 - what you already know before making calculation

<https://www.youtube.com/watch?v=rATNoxKg1yA>

https://www.youtube.com/watch?v=rATNoxKg1yA&ab_channel=JamesGilbert

EXAMPLE

H_A: artists typically tend to be Aries or Cancer

H₀:

Category	Observed	Expected
Aries	29	21.333
Taurus	24	21.333
Gemini	22	21.333
Cancer	19	21.333
Leo	21	21.333
Virgo	18	21.333
Libra	19	21.333
Scorpio	20	21.333
Sagittarius	23	21.333
Capricorn	18	21.333
Aquarius	20	21.333
Pisces	23	21.333

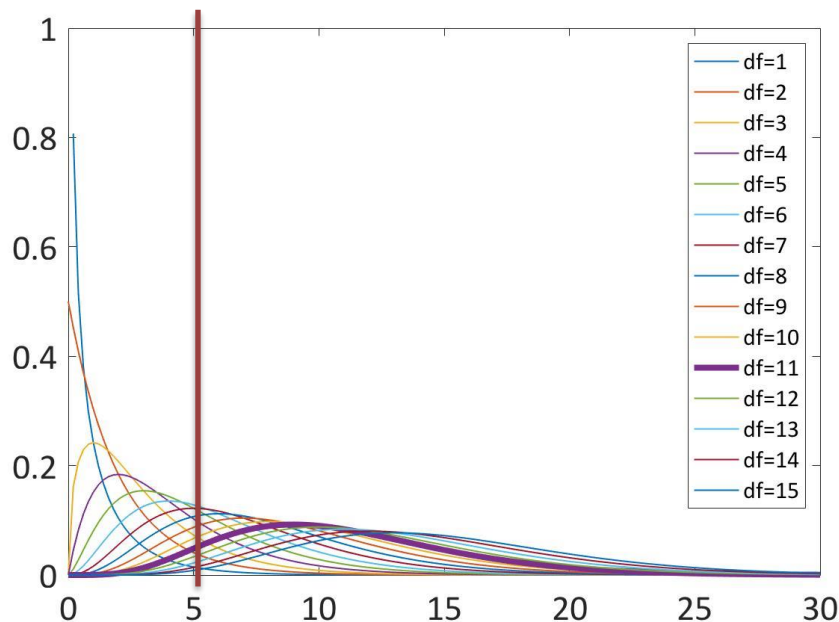
df = ?



256 artists



Please submit on
Moodle [12-Mar-
2025]:
Is the Null Hypothesis
Accepted or Rejected
based on Chi-Square
Test



Chi-square Distribution Table

d.f.	.995	.99	.975	.95	.9	.1	.05	.025
1	0.00	0.00	0.00	0.00	0.02	2.71	3.84	5.02
2	0.01	0.02	0.05	0.10	0.21	4.61	5.99	7.38
3	0.07	0.11	0.22	0.35	0.58	6.25	7.81	9.35
4	0.21	0.30	0.48	0.71	1.06	7.78	9.49	11.14
5	0.41	0.55	0.83	1.15	1.61	9.24	11.07	12.83
6	0.68	0.87	1.24	1.64	2.20	10.64	12.59	14.45
7	0.99	1.24	1.69	2.17	2.83	12.02	14.07	16.01
8	1.34	1.65	2.18	2.73	3.49	13.36	15.51	17.53
9	1.73	2.09	2.70	3.33	4.17	14.68	16.92	19.02
10	2.16	2.56	3.25	3.94	4.87	15.99	18.31	20.48
11	2.60	3.05	3.82	4.57	5.58	17.28	19.68	21.92
12	3.07	3.57	4.40	5.23	6.30	18.55	21.03	23.34
13	3.57	4.11	5.01	5.89	7.04	19.81	22.36	24.74
14	4.07	4.66	5.63	6.57	7.79	21.06	23.68	26.12

zodiac signs are evenly
distributed across artists

H_0

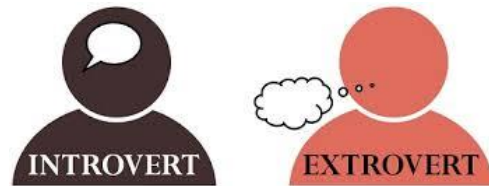
ACCEPTED



Chi-Square test for independence

test *relationship* between two separate variables

H_{01} = there is no relationship between extroversion and comfort level of dancing in public



test *difference* between two conditions

H_{02} = there is no difference in comfort level of dancing in public between introverts and extraverts

Chi-Square (example)

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

	Extroverts	Introverts	TOTAL
Not comfortable	10	40	50
comfortable	40	10	50
TOTAL	50	50	100

observed frequencies of Introverts and Extroverts who say they would or would not feel comfortable dancing in public

Chi-Square (example)

	Extroverts	Introverts	TOTAL
Not comfortable	10	40	50
comfortable	40	10	50
TOTAL	50	50	100

expected frequencies if the null hypothesis were true?

Chi-Square (example)

$$\frac{\text{row total} \times \text{column total}}{\text{total } n \text{ for table}}$$

	Extroverts	Introverts	TOTAL
Not comfortable	10 25	40 25	50
comfortable	40 25	10 25	50
TOTAL	50	50	100

observed and **expected** frequencies Introverts and Extroverts who say they would or would not feel comfortable dancing in public

Chi-Square (example)

	Extroverts	Introverts	TOTAL
Comfortable	10	40	50
Not Comfortable	40	10	50
TOTAL	50	50	100

$$\chi^2 = \sum \frac{(\text{observed} - \text{expected})^2}{\text{expected}}$$

$$\frac{(40 - 25)^2}{25} + \frac{(10 - 25)^2}{25} + \frac{(10 - 25)^2}{25} + \frac{(40 - 25)^2}{25}$$

9 + 9 + 9 + 9

$$\chi^2 = 36$$

df=?

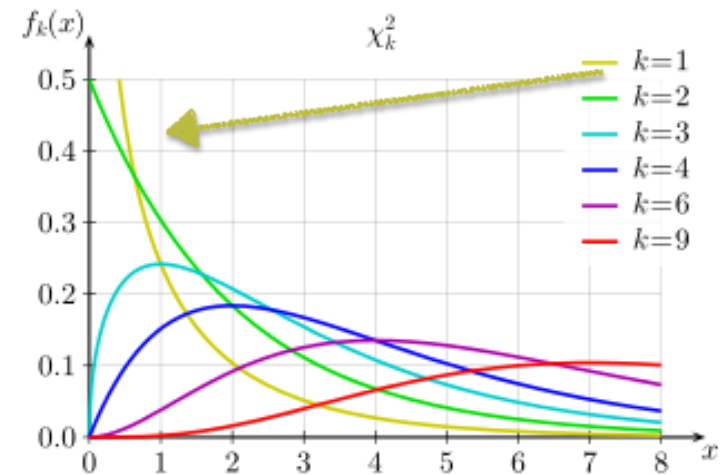
Chi-Square (example)

- Degrees of freedom for independence
 - number of cells you would need to calculate all other cell values, assuming we know marginal values

$$df = (R-1)(C-1)$$

$$df = (2-1)(2-1) = 1$$

- Our chi-square is significant
 - Introverts tend to feel more comfortable dancing in public compared to Extroverts (surprise!)
 - Cooked-up data!!

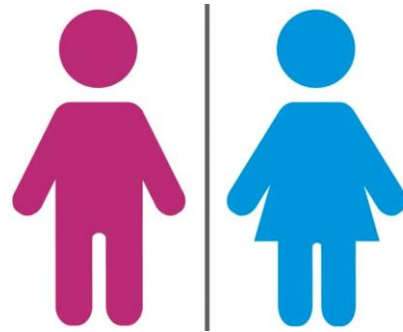


$$\chi^2 = 36$$

H_0 **REJECTED**

Gender Study

EXAMPLE



MENTAL
HEALTH
FIRST AID®

H₀: There is no relationship between gender and willingness to use mental health services

H_{A1}: The distribution of reported willingness to use mental health services for males has proportions that are different from those in females

H₀: The distribution of reported willingness to use mental health services has the same proportions for males and females

H_A: The distribution of reported willingness to use mental health services for males has proportions that are different from those in females

Contingency Table

row total × column total
total n for table

Willingness to Use Mental Health Services (n=150)

	No	Maybe	Yes	Total
Males	17 12	32 30	11 18	60
Female	13 18	43 45	34 27	90
Total	30	75	45	150

$$df = (R-1)(C-1)$$

$$df = 2$$

Note: How did we estimate “expected” values?

30 said NO - so we would expect equal distribution of 15 and 15.

BUT only 40% are females so 40% of 30 makes it 12.

Gender Study

EXAMPLE

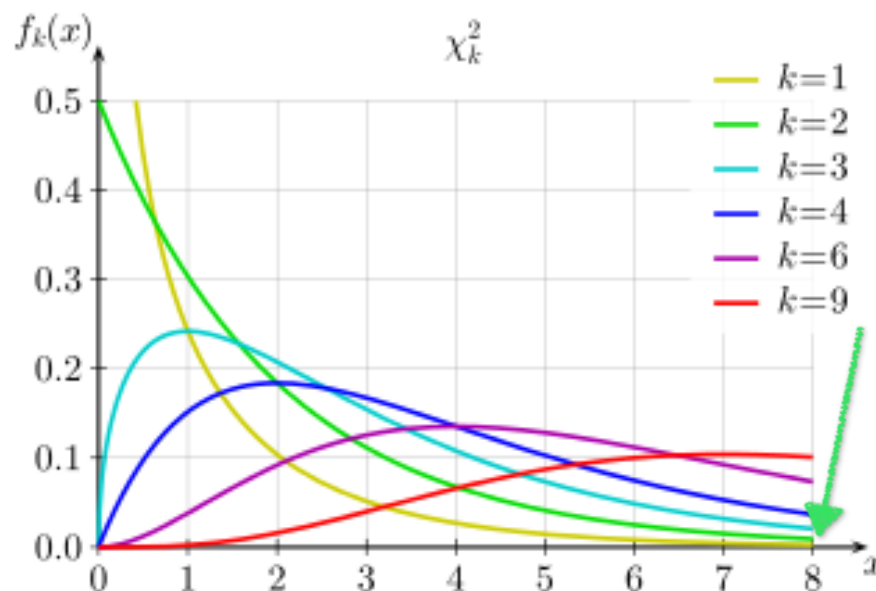
Willingness to Use Mental Health Services

$$\chi^2 = 8.23$$

$$df = 2$$

H_0

REJECTED



Males are less willing to use Mental Health Services

Effect Size

Effect size in Chi square

- For a 2×2 table $\rightarrow$ Phi Coefficient

$$\phi = \sqrt{\frac{\chi^2}{n}}$$

$\rightarrow$ Correlation between two categorical variables

Phi of 0.1 small, 0.3 medium, 0.5 large

- For larger tables $\rightarrow$ Cramer's V coefficient ($> 2 \times 2$)

$$V = \sqrt{\frac{\chi^2}{n \times df^*}}$$

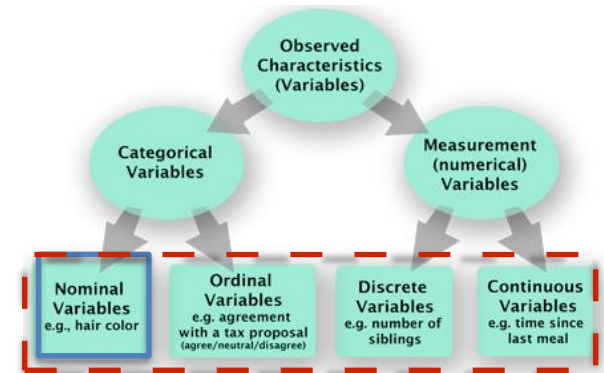
$\rightarrow$ Df^* is the smallest of $C-1$, $R-1$

$$\text{Phi} = \sqrt{8.23/150} \\ = 0.23$$

Results showed a significant difference between males' and females' attitude toward using mental health services,
 $\chi^2 (2, n = 150) = 8.23, p < .05, V = 0.23$

Chi-Square Test and Correlation

Participant	Self-Esteem X	Academic Performance Y
A	13	73
B	19	88
C	10	71
D	22	96
E	20	90
F	15	82
.	.	.
.	.	.
.	.	.



		Level of Self-Esteem			
		High	Medium	Low	
Academic Performance	High	17	32	11	60
	Low	13	43	34	90
		30	75	45	$n = 150$

Chi-square and independent measures t and ANOVA

Participant	Self-Esteem X	Academic Performance Y
A	13	73
B	19	88
C	10	71
D	22	96
E	20	90
F	15	82
.	.	.
.	.	.
.	.	.

		Level of Self-Esteem			
		High	Medium	Low	
Academic Performance	High	17	32	11	60
	Low	13	43	34	90
		30	75	45	$n = 150$

Median Test for Independent Samples

- non-parametric alternative to independent measures *t*-test (or ANOVA) to determine significant group differences
- H_0 = different samples come from population that share a common median

Self-Esteem Scores for Children at Three Levels of Academic Performance							
High		Medium				Low	
22	14	22	13	24	20	11	19
19	18	18	22	10	16	13	15
12	21	19	15	14	19	20	16
20	18	11	18	11	10	10	18
23	20	12	19	15	12	15	11

Median Test for Independent Samples

- calculate median for combined group ($n = 40$)
- within each group, perform median (17) split and fill contingency table

Self-Esteem Scores for Children
at Three Levels of Academic Performance

High		Medium				Low	
22	14	22	13	24	20	11	19
19	18	18	22	10	16	13	15
12	21	19	15	14	19	20	16
20	18	11	18	11	10	10	18
23	20	12	19	15	12	15	11

	Academic Performance		
	High	Medium	Low
Above Median	8	9	3
Below Median	2	11	7

Median Test for Independent Samples

	Academic Performance		
	High	Medium	Low
Above Median	8 5	9 10	3 5
Below Median	2 5	11 10	7 5

$$\chi^2 = 5.4 \quad df = 2 \quad \chi^2 = 5.99 (p < .05)$$

—> not sufficient evidence to conclude that there are significant differences among the self-esteem for these three groups of students

Chi-Square test

Limitations

- Observations must be unique to one cell (Between subjects)
 - each person must fall into only one cell
 - not valid for within subject designs (repeated measures/matched pairs)
- Only frequencies can be studied, not means, percentages, ratios, etc.
- Low **expected** frequencies cause problems (should be ≥ 5)
 - loss of statistical power
- No group should contain less than 10 (or 5) (try to regroup instead)
- Not apt for low sample size.
- Informs of presence or absence (probability of occurrence) of association but doesn't measure strength of association

Life after chi-squared: an introduction to log-linear analysis.

Streiner DL¹, Lin E.

Author information

- 1 Kunin-Lunenfeld Applied Research Unit, Baycrest Centre for Geriatric Care, North York, Ontario. dstreiner@rotman-baycrest.on.ca

Abstract

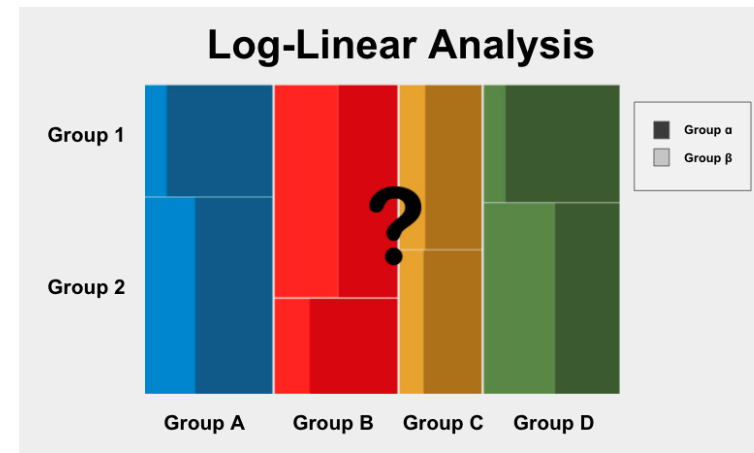
Chi-squared tests are used to examine the relationships among categorical variables. However, they are difficult to use and interpret when more than 2 variables are involved. In such cases, it is better to use a related statistic, called log-linear analysis. This article is an introduction to log-linear models, illustrating how they can be used to tease apart relationships among several variables in looking at the factors associated with photonumerophobia.

	Age category of car		
	New	Old	Total
Male drivers			
Behaviour at amber light			
Stopped	79	63	142
Did not stop	87	95	182
Total	166	158	324
Female drivers			
Behaviour at amber light			
Stopped	95	83	178
Did not stop	51	94	145
Total	146	177	323
Total old/new cars:	312	335	647

Table 18.15 Stopping behaviour of male and female drivers in old and new cars.

Log-Linear Analysis

- variable of interest is proportional or categorical
- have two or more options
- no assumptions of IV or DV
- used for both hypothesis testing and model building



Different types of tests (Summary)

Test type	Between subjects designs (Independent samples)	Within subject designs (repeated measures/matched pairs)
Non-parametric (for categorical data)	Chi-square	<i>The binomial sign test</i>
Non-parametric (for ordinal data)	<i>Mann-Whitney U</i>	<i>Wilcoxon Signed-Rank test</i> <i>The binomial sign test</i>
Parametric	<i>Unrelated t-test (level of data: interval)</i>	<i>Related t-test (level of data: interval)</i>